

Adam Torres Encarnacion

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EDUCATION

The Pennsylvania State University
M.S. in Artificial Intelligence

Expected: May 2028

The Pennsylvania State University
B.S. in Data Science, B.S. in Statistics

Expected: Dec 2026

GPA: 3.7

WORK EXPERIENCE

Amazon
Software Engineering Intern

May 2026 – Aug 2026
Cupertino, CA

- Refactored **C++ infrastructure** for 2 fiber-cleaning robot arms projected to save \$28.6M/year across 85 data centers
- Built a **19-route HTTP server** to control robot arms remotely over **SSH**, maintaining the local control loop **under 2ms**
- Made an **LLM-as-judge** classification pipeline for 6 optical fiber and transceiver labels, storing live detection data in **S3**
- Created a self-evolving MCP knowledge base for agent-first development, migrating **~400 legacy docs from 2 repos**
- Hardened 2 AWS CI/CD pipeline stages with blocking lint checks, code complexity checks, and **~40** integration tests

IBM
AI Engineering Intern

May 2025 – May 2026
Remote

- Shipped an LLM-powered **REST microservice** to power documentation for 3,100+ scripts, saving \$223,000 in labor costs
- Helped triage **~500k** criminal records spanning 20+ years with a **LangChain** chatbot built on Python, cutting review time
- Deployed serverless **Azure Function Apps** that autoscaled with incoming jobs, cutting manual review workload by **78%**
- Refined a file-chunking pipeline using Llama-4 and GPT-4o-mini to handle 7+ large-scale internal processing requests

ACHIEVEMENTS

- 1st Place at HopHacks, Tiger Data Track Sep 2026
- 1st Place at YHack, K2 Think V2 Track Mar 2026
- 2nd Place at NexHacks, ByteDance Track (Solo) Jan 2026
- 1st Place at HackPrinceton, xAI Track Nov 2025
- Awarded a Jane Street \$5,000 scholarship (AMP), 1 of 80 recipients nationally Jul 2023

PROJECTS

[🔗 Agent-Driven Vision Annotation \(OpenAI WebMCP Challenge '26\)](#) | *Python, WebMCP, OpenCV*

- Turned AI agents into click-free CV data annotators through **26 WebMCP tools** to draw & navigate an open canvas
- Built an OpenCV vision keyboard emulator with **98.78% F1** on 1,253 AI-labeled images after 600 MPS-accelerated epochs
- Balanced image datasets using 8 deterministic image transforms and GPT-Image-2 in asynchronous augmentation jobs
- Supported **16 concurrent labeling sessions** at <300 ms p95 metadata API latency using FastAPI, PostgreSQL, and **S3**

[🔗 Multimodal Drawing Canvas \(NexHacks '26 ByteDance 2nd place\)](#) | *TypeScript, Gemini, PostgreSQL*

- Won 2nd (out of 1500+) **solo** with a tldraw canvas that auto-completes sketches in place via Gemini AI image generation
- Used Gemini to **classify** chat vs. drawing requests, then cropped subjects with **Grounding DINO** on **CUDA** (PyTorch)
- Integrated OpenAI Whisper for voice prompts and gated each image generation behind an on-canvas accept/reject overlay
- Saved tldraw snapshots as **PostgreSQL JSONB** on a 2s debounce, generating PNG previews and skipping saves offline

[🔗 Multi-Agent Social Simulation \(YHack '26 K2 Think V2 1st place\)](#) | *Python, LangGraph, asyncio*

- Built a multi-agent simulation engine with FastAPI and **LangGraph StateGraph**, using asyncio for concurrent agent execution, shared memory streams, and Pydantic validation for structured outputs
- Modeled echo-chamber formation across **50+ agents** using recency-weighted memory retrieval, threshold-triggered reflection, and Deffuant bounded-confidence opinion updates; streamed simulation updates over Socket.IO

TECHNICAL SKILLS

Programming Languages: Python, C++, C, C#, SQL, TypeScript, HTML/CSS, R, Lua

Frameworks: PyTorch, OpenCV, Hugging Face, Pandas, NumPy, FastAPI, React, Node.js, Next.js, LangChain/LangGraph

Developer Tools: Git, Linux, VS Code, Vim/Neovim, Pi (harness), Cmux (terminal multiplexer), Agile/Scrum

Databases: PostgreSQL, MySQL, SQLite, Redis, MongoDB, DynamoDB

Cloud/Infra: AWS (Lambda, S3, EC2, Bedrock, DynamoDB), Azure Functions, Vercel, Docker, CI/CD (GitHub Actions)